IMPACT OF HOT AND WET GRANULATION TECHNIQUES ON COMPRITOL® 888 ATO LIPID MATRICES FOR EXTENDED RELEASE OF HIGHLY WATER SOLUBLE HIGH DOSE API

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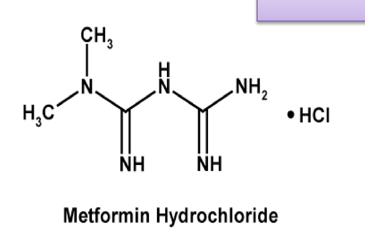
Poster No. 363

INTRODUCTION

Metformin Hydrochloride



- An oral antihyperglycemic.
- Freely soluble in water,
- Belongs to BCS Class 3
- pKa of 12.4.

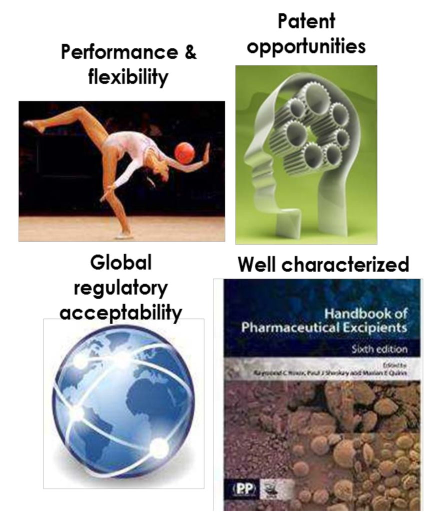
- Has a half-life of 6 hours with peak plasma concentrations (Cmax) achieved within one to three hours of taking immediate-release Metformin and four to eight hours with extended-release formulations.
- API possesses poor compaction characteristics and is hygroscopic in nature thereby rendering it difficult to process on an industrial level.

Compritol® 888 ATO: A versatile lipid matrix forming agent

Chemical composition:
Mixture of mono-, di- and tribehenate (C22) of glycerol

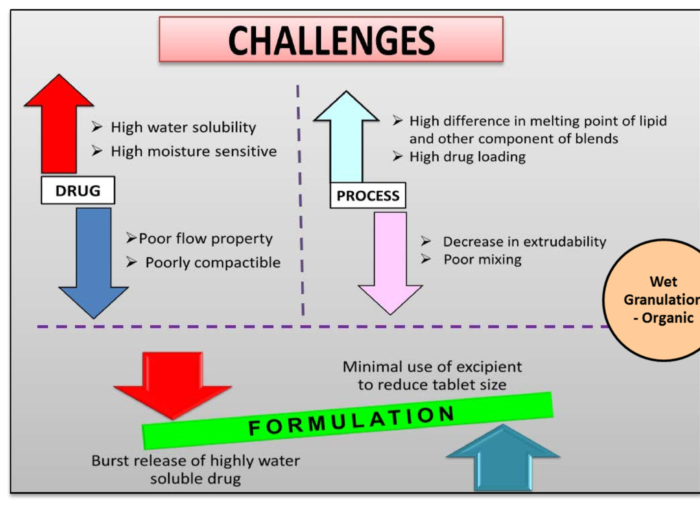
Physical properties:
- Fine, white powder of 50 µm
- Spherical particles
- Drop point 69 – 74°C
- HLB 1, water insoluble
- Stable over 3 years at < 35°C

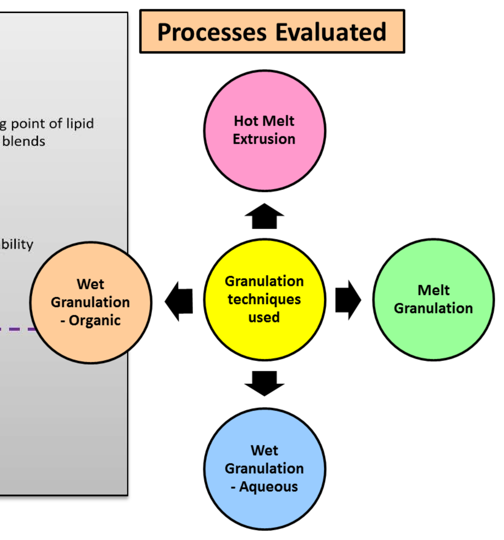
Regulatory information:
- Conform to USP/NF/EP/Chinese Ph.
- GRAS, FDA IIG, List of Acceptable Non-medicinal Ingredient.



OBJECTIVE

To evaluate the impact of different granulation techniques on lipid matrices formed by Compritol® 888 ATO for retarding the release of Metformin Hydrochloride Extended Release Tablets (MHERT) 500 mg.

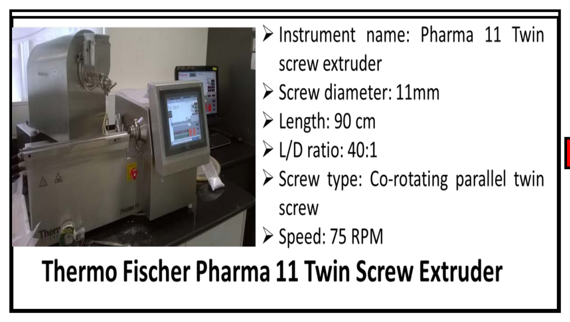


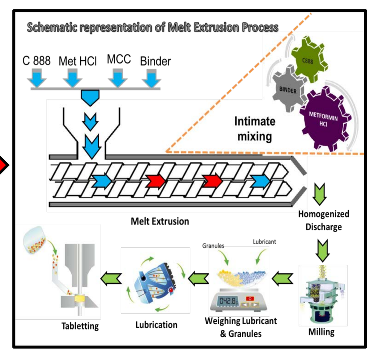


Role of Ingredients	Granulation Process			
	HME	HMG	Wet aqueous	Wet organic
API	Metformin Hydrochloride			
Release Modifier	Compritol® 888 ATO			
Binder	Hydroxy Propyl Cellulose		PVP K30	
Diluent	Avicel® PH 101 (Intragranular) Avicel® PH 200 (Extragranular)			
Extragranular adsorbent	Aerosil® 200			
Glidant	Syloid® 244 FP			
Lubricant	Magnesium Stearate			

EXPERIMENTAL

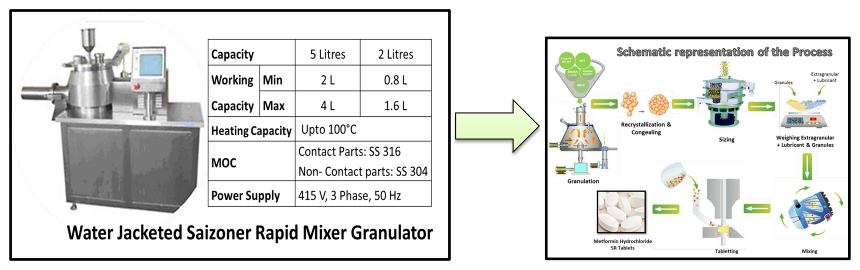
Hot Melt Extrusion Process





Ingredients	Drug loading (%)							
	F1	F2	F5	F8	F9	F10	F11	F12
Metformin HCl	65	65	65	65	75	75	75	75
Compritol® 888 ATO	25	35	30	25	20	15	10	5
Starch 1500	10	-	-	-	10	-	-	-
Klucel® LF	-	10	-	5	-	10	-	5
Avicel® PH 101	-	-	10	5	-	-	10	5
Temperature (°C)	90	90	100	100	150	120	130	150
RPM	75	75	75	75	75	75	75	75
Torque (%)	30 - 60							

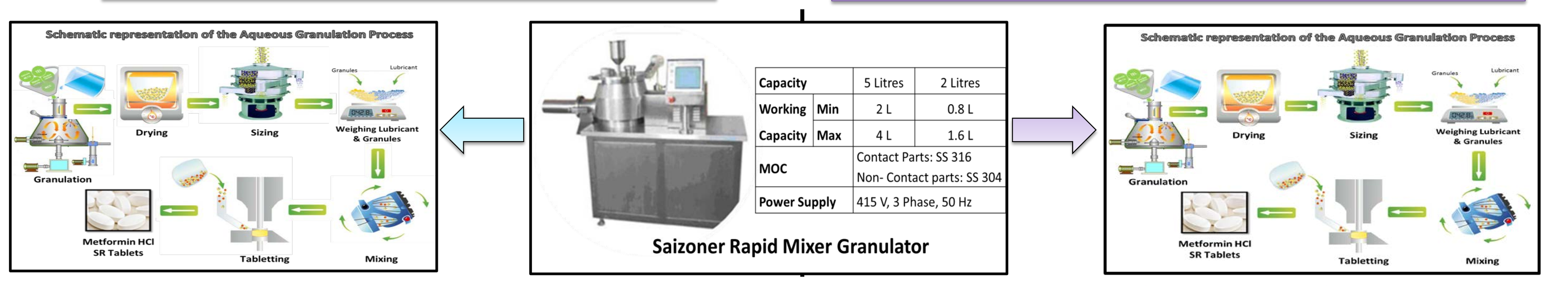
Hot Melt Granulation Process



EXCIPIENTS	% in formula	mg / Tablet
INTRAGRANULAR EXCIPIENTS		
Metformin Hydrochloride	62.5	500
Compritol® 888 ATO	25	200
Klucel® LF	6	48
Avicel® PH 101	6.5	52
EXTRAGRANULAR EXCIPIENTS		
Avicel® PH 200	3.53	30
Aerosil® 200	2	16
Magnesium Stearate	0.5	4
Total Tablet Weight		850

Wet Aqueous Granulation Process

Wet Non – Aqueous Granulation Process



Capacity		5 Litres	2 Litres
Working Capacity	Min	2 L	0.8 L
	Max	4 L	1.6 L
MOC		Contact Parts: SS 316 Non- Contact parts: SS 304	
Power Supply		415 V, 3 Phase, 50 Hz	

Wet Aqueous Granulation:

Intragranular	% in formula	mg / Tablet
Intragranular Excipients		
Metformin Hydrochloride	48.31	500
Compritol 888 ATO	38	393.3
Plasdone™ K 29/32	5	51.75
Avicel® PH 101	4.94	51.129
Extragranular Excipients		
Prosolv® SMCC 90 HD	3	31.05
Syloid® 244 FP	0.25	2.5875
Magnesium Stearate	0.5	5.175
Total Tablet weight (mg)		1035

Wet Non – Aqueous Granulation:

Intragranular	% in formula	mg / Tablet
Intragranular Excipients		
Metformin Hydrochloride	52.63	500
Compritol 888 ATO	35	332.5
Plasdone™ K 29/32	3	28.5
Avicel® PH 101	5.26	49.97
Extragranular Excipients		
Prosolv® SMCC 90 HD	3.43	32.585
Syloid® 244 FP	0.25	2.375
Magnesium Stearate	0.5	4.75
Total Tablet weight (mg)		950

Granulating Solvent: Methylene Chloride

Evaluation

Physical Characterization
- Bulk density, Tap density & Flow properties of granules using Electrolab Densitometer
- Avg. weight, Hardness and thickness of tablets.

Dissolution Studies
- Test 1 & 2
- Apparatus: 2 (Paddle Type)
- Phosphate buffer, pH 6.8, 1000 ml
- 100 rpm
- Time points: 1, 2, 3, 6 & 10 hours

Differential Scanning Calorimetric, DSC
- Heat rate: 35°C to 300°C @ 10°C / min.
- Pyris 6, Perkin Elmer.

Scanning Electron Microscopy
- Performed using JSM-5200 (JEOL, Japan) operated at 20 KV with Platinum sputtering for 20 sec
- samples mounted on carbon tape

Content of Methylene Chloride by Gas Chromatography
- USP 467 monograph on Residual solvents as specified in USP 32/ NF 27
- Identification, quantification & control of Class 2 Residual solvent in water soluble substances (Procedure C) by head space analysis.

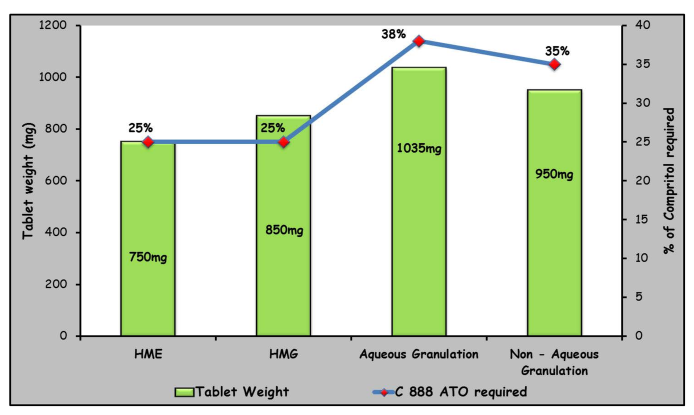
RESULTS AND DISCUSSION

Physical Characterisation

USP Specification			Observation				
Compressibility Index (%)	Flow Characteristics	Hausner Ratio		HME	HMG	Aqueous	Non - aqueous
< 10	Excellent	1.00 - 1.11	Bulk Density (g/ml)	0.513	0.57	0.41	0.4
11 - 15	Good	1.12 - 1.18	Tap Density (g/ml)	0.645	0.80	0.51	0.65
16 - 20	Fair	1.19 - 1.25	Compressibility Index (%)	20.51	28.57	20.41	38.00
21 - 25	Passable	1.26 - 1.34	Hausners ratio	1.258	1.4	1.26	1.61
26 - 31	Poor	1.35 - 1.45					
32 - 37	Very Poor	1.46 - 1.59					
> 38	Very very Poor	> 1.60					

- ✓ Granules prepared using HME process showed appreciable flow & compaction properties
- ✓ Granules prepared using Non – Aqueous granulation processes exhibited poor flow and compression properties with prominent tablet defects.
- ✓ Flow properties of granules prepared using HMG & aqueous granulation processes were passable and afforded good tablets after addition of extragranular material like MCC & Silicon dioxide.

Tablet Parameters	HME	HMG	Aqueous	Non – aqueous
Description	White, Oval shaped tablets	White, Oval shaped tablets	White, Capsule shaped tablets (Caplets)	White, Capsule shaped tablets (Caplets)
Dimensions	Length: 16 mm Width: 9 mm Thickness: 7 mm	Length: 16 mm Width: 9 mm Thickness: 8 mm	Length: 19 mm Width: 9 mm Thickness: 6.8 mm	Length: 19 mm Width: 9 mm Thickness: 5.95 mm
Hardness range (Avg.10 tablets)	80 – 120 N	80 – 120 N	134.9 – 142.5 N	100 – 120 N

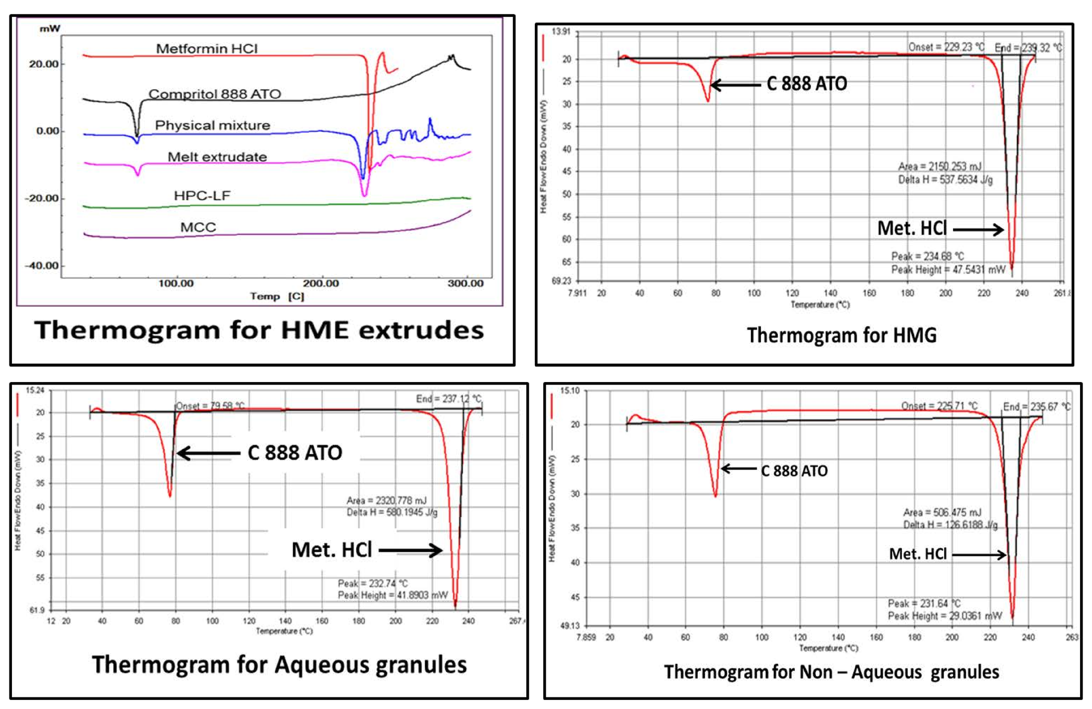


Plot indicating % of Compritol 888 ATO required and its impact on Tablet weight

Melt processes:
- ✓ Desired release profile achieved at 25% Compritol 888 ATO levels in formulation.
- ✓ Optimized formulation with low tablet weight: 750 mg for HME & 850 mg for HMG.
- ✓ Minimum extragranular excipients required to achieve desired release profile

Wet Granulation processes:
- ✓ Desired release profile achieved at 38% Compritol 888 ATO levels for aqueous granulation & at 35% for Non - aqueous granulation .
- ✓ Higher tablet weights in comparison to melt processes.

Differential Scanning Calorimetry



Thermogram for HME extrudes; Thermogram for HMG; Thermogram for Aqueous granules; Thermogram for Non – Aqueous granules

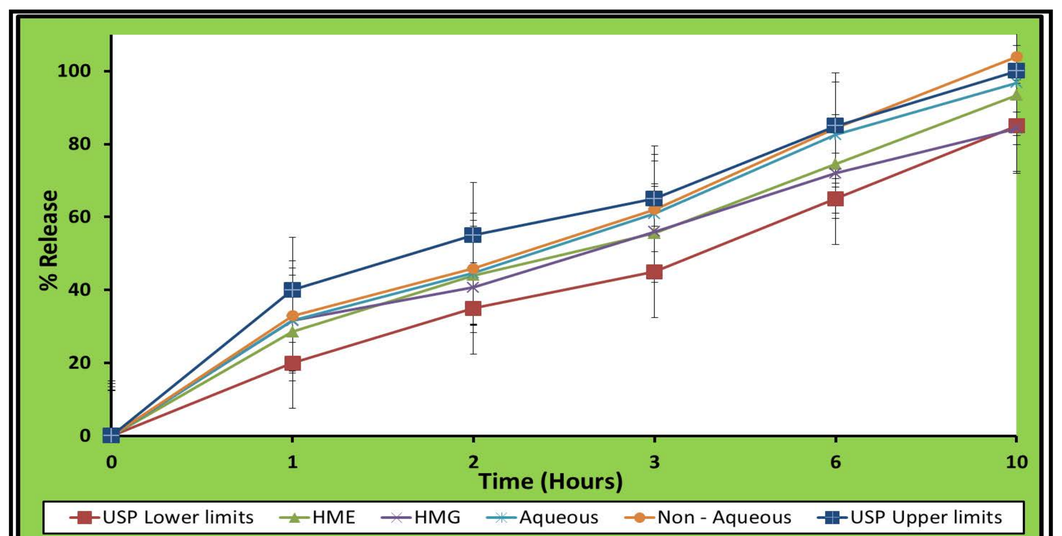
- ✓ Hot melt processes viz. extrusion & granulation resulted in formation of partial solid dispersion of Metformin with Compritol 888 ATO as a decrease in endotherm for Compritol is observed in both processes.
- ✓ No change in the endotherms of Metformin HCl & Compritol 888 ATO were observed in case of aqueous & non aqueous granulation processes indicating formation of physical mixture bound by Polyvinly pyrrolidone as binder.

Content of Methylene Chloride by Gas Chromatography

Class of Solvent	Concentration Limit as per USP 32 / NF 27	Observed Value
Class 2	600 ppm	486 ppm

- ✓ The content of Methylene Chloride in the dried granules prepared using DCM as solvent was well within specification.

Dissolution Studies

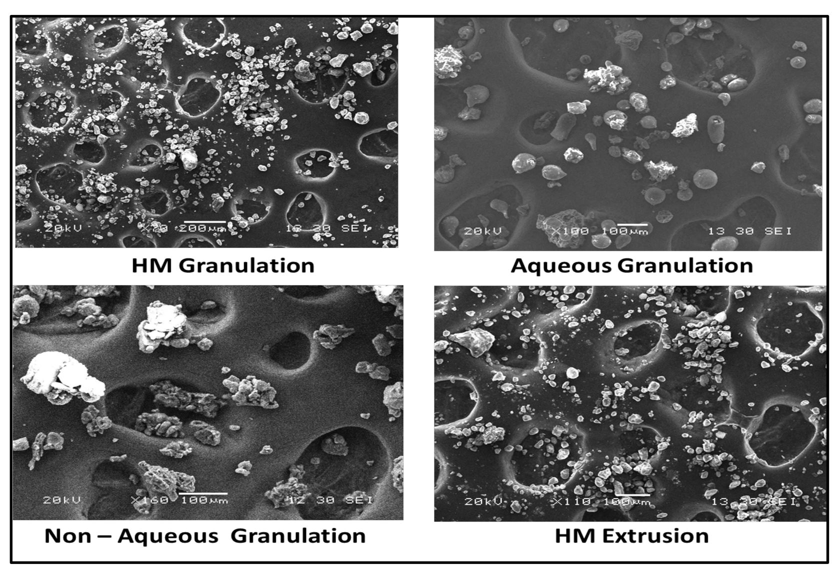


Plot of % Release versus Time for Metformin HCl SR Tablets prepared using HME, HMG, Aqueous and Non – Aqueous granulation process

Time (Hours)	Amount Dissolved (500 mg)
1	20 - 40 %
2	35 - 55 %
3	45 - 65 %
6	65 - 85 %
10	NLT 85 %

Dissolution profiles of Metformin HCl Extended Release tablets, 500 mg prepared using Hot Melt Extrusion, Hot Melt Granulation, Aqueous Granulation and Non Aqueous Granulation techniques complies with the specifications of Test 1 & 2 for dissolution of Metformin HCl Extended Release tablets, USP 37 / NF 32.

Scanning Electron Microscopy



- ✓ In case of Hot melt processes intimate mixing of Metformin HCl and Compritol® 888 ATO is observed resulting in formation of partial solid dispersion.
- ✓ Stability of partial solid dispersion can be attributed to the addition of Hydroxy Propyl Cellulose (Klucel LF) as meltable binder.
- ✓ In case of aqueous & non – aqueous wet granulation processes Metformin HCl & Compritol® 888 ATO co – exist as physical mixture bound together by Polyvinly pyrrolidone (Plasdone® K 29/32) as binder.

Summary

- ✓ Hydroxy Propyl Cellulose (Klucel LF) as meltable binder proved effective in Hot Melt processes (HME & HMG) while Polyvinyl Pyrrolidone (Plasdone™ K29 /30) proved effective as binder for wet granulation processes (Aqueous & Non – aqueous).
- ✓ Flow properties of the granules can be graded as: HME > HMG ≡ Aqueous granulation > Non – Aqueous granulation.
- ✓ Tablet weights of Metformin HCl Extended Release tablets, 500 mg were observed to be: HME (750 mg) < HMG (850 mg) < Non – aqueous granulation (950 mg) < Aqueous granulation (1035 mg).
- ✓ Hot melt processes resulted in slight fusion of Metformin HCl and Compritol® 888 leading to formation of partial solid dispersion with the aid of Klucel LF, while aqueous & non – aqueous wet granulation processes Metformin HCl & Compritol® 888 ATO co – exist as physical mixture bound together by Polyvinly pyrrolidone (Plasdone™ K 29/32) as binder.
- ✓ Optimized formulations prepared using the four granulation techniques gave tablets that complied with dissolution Test 1 & 2 specified in the USP 37/ NF 32 monograph for Metformin HCl Extended Release Tablets, 500 mg.

Conclusion

Amongst the four granulation processes evaluated the melt processes provided Metformin HCl Extended release tablets USP/NF, 500 mg with low tablet weight, lesser quantity of excipients and ease of processability.

Acknowledgement

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